

Quiz 03

Total points 28/30

1. Quiz will begin at 12pm and will end at 12:30pm on 18.03.2021.
2. You are required to submit google form by 12:30 PM. Unsubmitted responses will be given zero marks automatically.
3. Keep your video on, through google meet, while taking quiz.
4. Make sure you have a working desktop/laptop.
5. All responses are required otherwise google form won't be submitted. There will be no negative marking.

✓ Why do we perform rotation for receiver function's calculation? * 3/3

The two horizontal components are rotated to the radial and tangential directions (N-s to radial and E-W to transverse). So that the effective source time function of each earthquake is obtained by stacking all vertical components recorded by the network, after aligning them at the first P arrival.

✓ Which technique of receiver functions is famously known as Kanamori 2/2 technique? *

- ☒ H-K stacking
- ☐ CCP stacking
- ☐ Iterative technique
- ☐ None of the above



✓ Draw a ray path diagram for two layered flat earth for Ps, PpPs and PsPs+PpSs phases. * 3/3

 amit bajpai - Ami...



✓ Define receiver functions. *

3/3

Receiver functions are time series, computed from three-component seismograms after the deconvolution of vertical component from the radial component, which show the relative response of Earth structure near the receiver.

✓ What are different types of receiver functions? *

2/2

1) Ps receiver function (P-to S converted phases) (2) SP receiver function (S-to-P converted phases)



True/False *

	True	False	Score	
CCP stacking of receiver functions involves two way travel time of teleseismic earthquakes.	☐	☒	2/2	✓
Receiver functions are generally computed from teleseismic earthquake data.	☒	☐	2/2	✓
Receiver functions generally have PpSs+PsPs type of negative polarity phase.	☒	☐	2/2	✓
Generally, in H-K stacking, least weight is given to Ps phase of receiver function.	☐	☒	2/2	✓
Water level deconvolution is done in frequency domain.	☒	☐	2/2	✓



✓ Briefly describe the process of iterative time domain deconvolution. * 3/3

1) Get the observed vertical component and observed radial component (2) Assume the initial receiver function and convolve with observed vertical component (3) then by convolution we will get calculated radial function (4) calculate the difference between calculated radial function and observed one (5) repeat the process by changing the receiver function, we can add more and more peaks (6) The iteration for which the difference is minimum that corresponds to the more accurate receiver function

✗ Which of the following technique is a grid search technique? * 0/2

- ☐ H-K stacking
- ☒ CCP stacking
- ☐ Monte carlo inversions
- ☐ None of the above

✗

Correct answer

- ☒ H-K stacking

✓ What is the full form of CCP stacking? 2/2

Common Conversion Point

✓

This form was created inside of Indian Institute of Technology (Indian School of Mines), Dhanbad.

Google Forms

